Lily's Pad - Prototypage

MODULE ELEC 06/26 - 10/26

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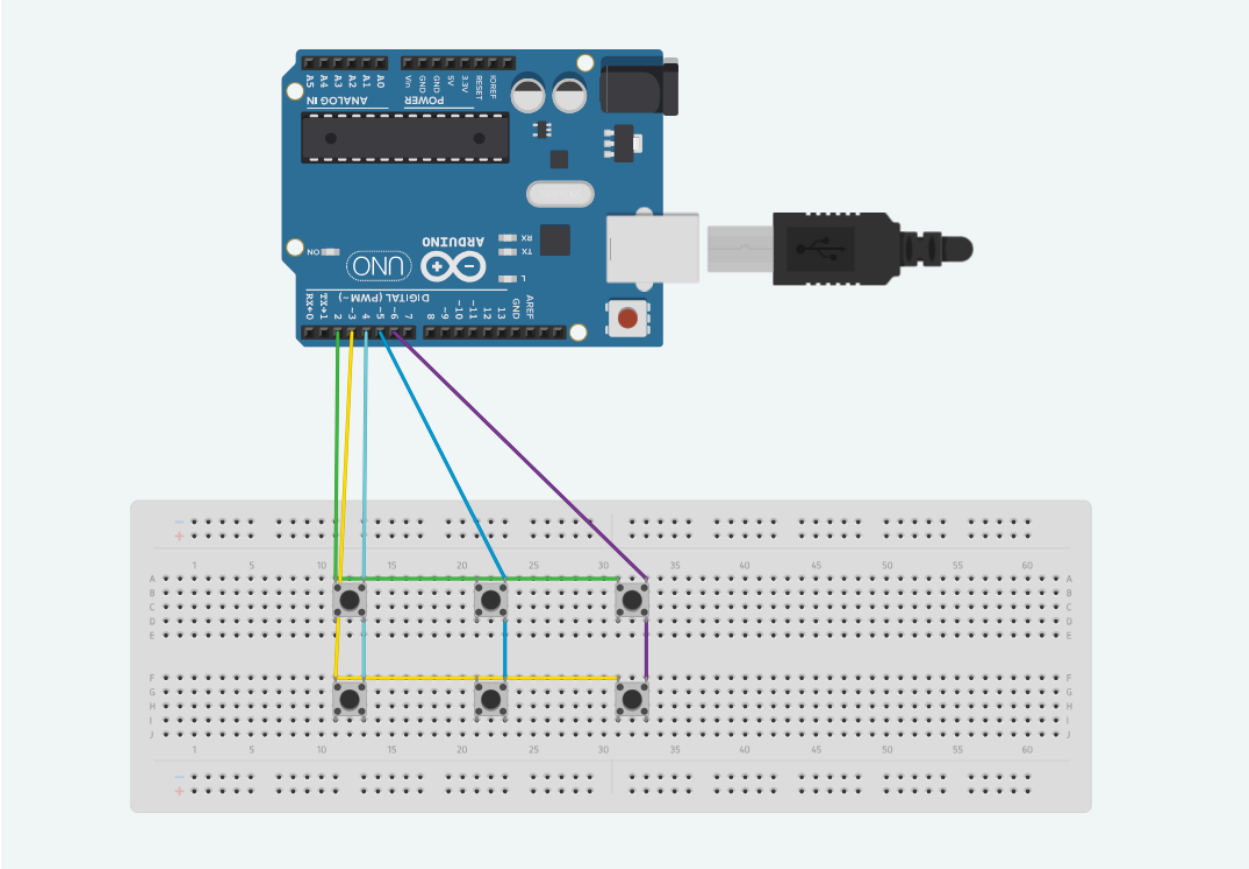
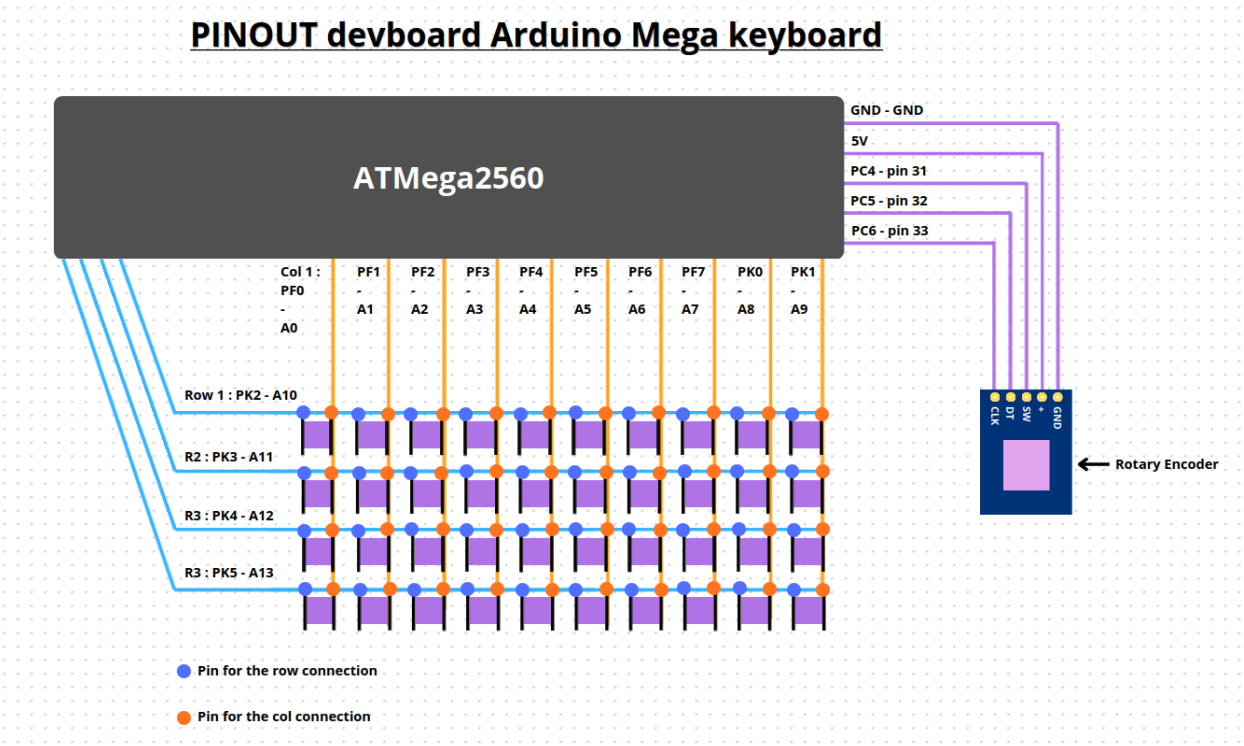
KEYBOARD

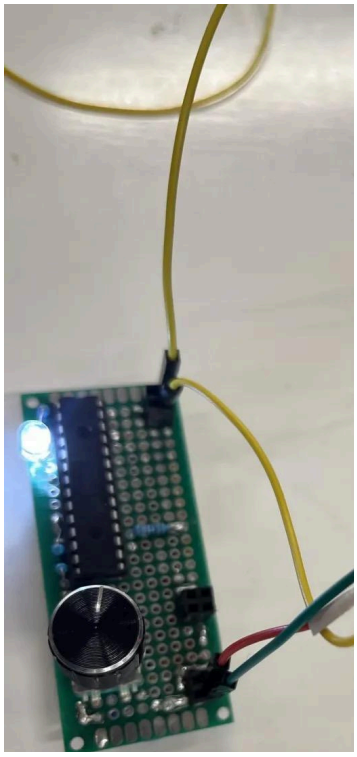


11/06/26

First test for the keyboard, first 3x3 matrix to start trying to make it work without getting lost with the 40 buttons. Diagram created on Tinkercad.

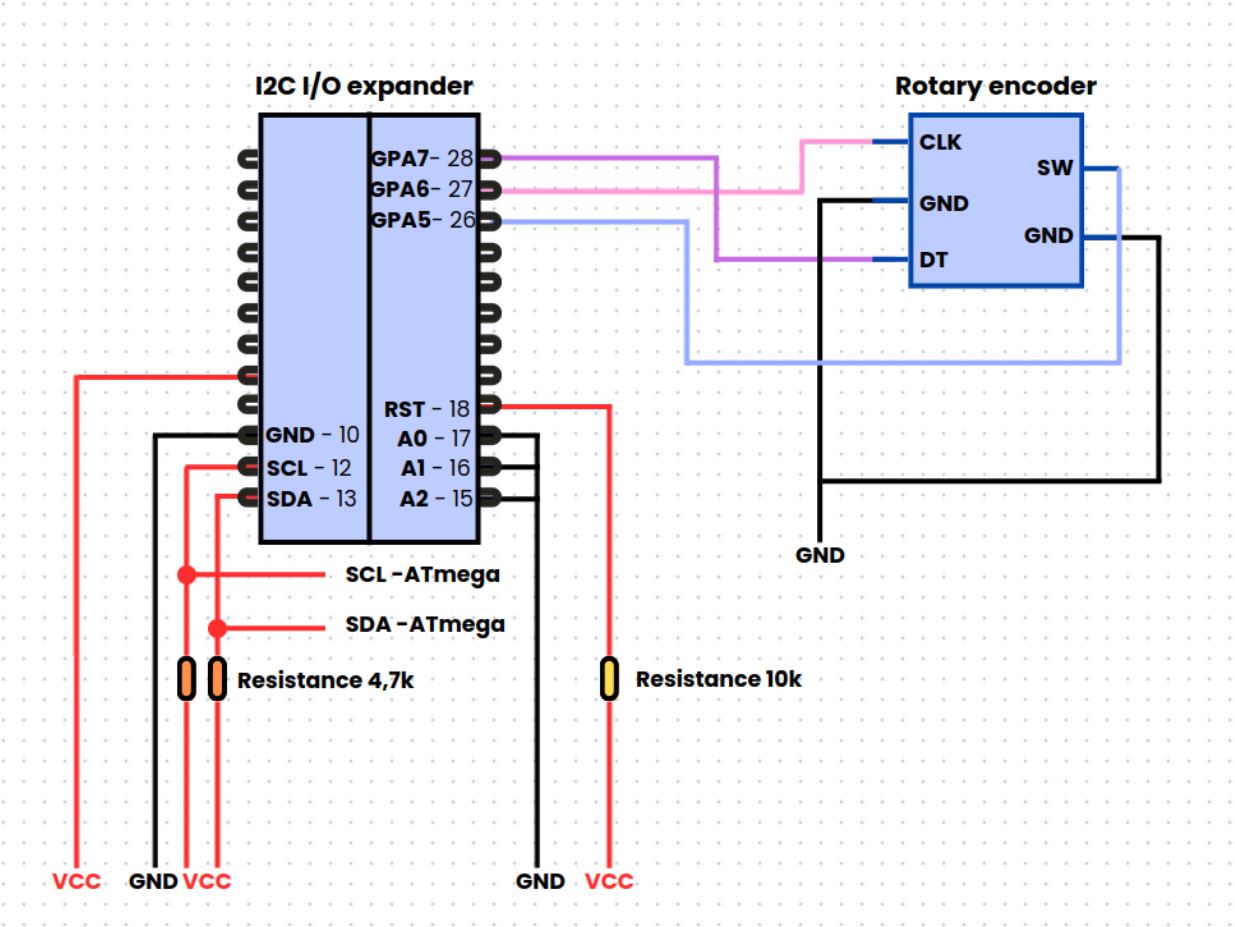


Functional connection of the keyboard and the rotary encoder, a code that makes everything work but needs to be refined. Here, the rotary is still in GPIO; you have to switch it to I2c and add the slide switch.

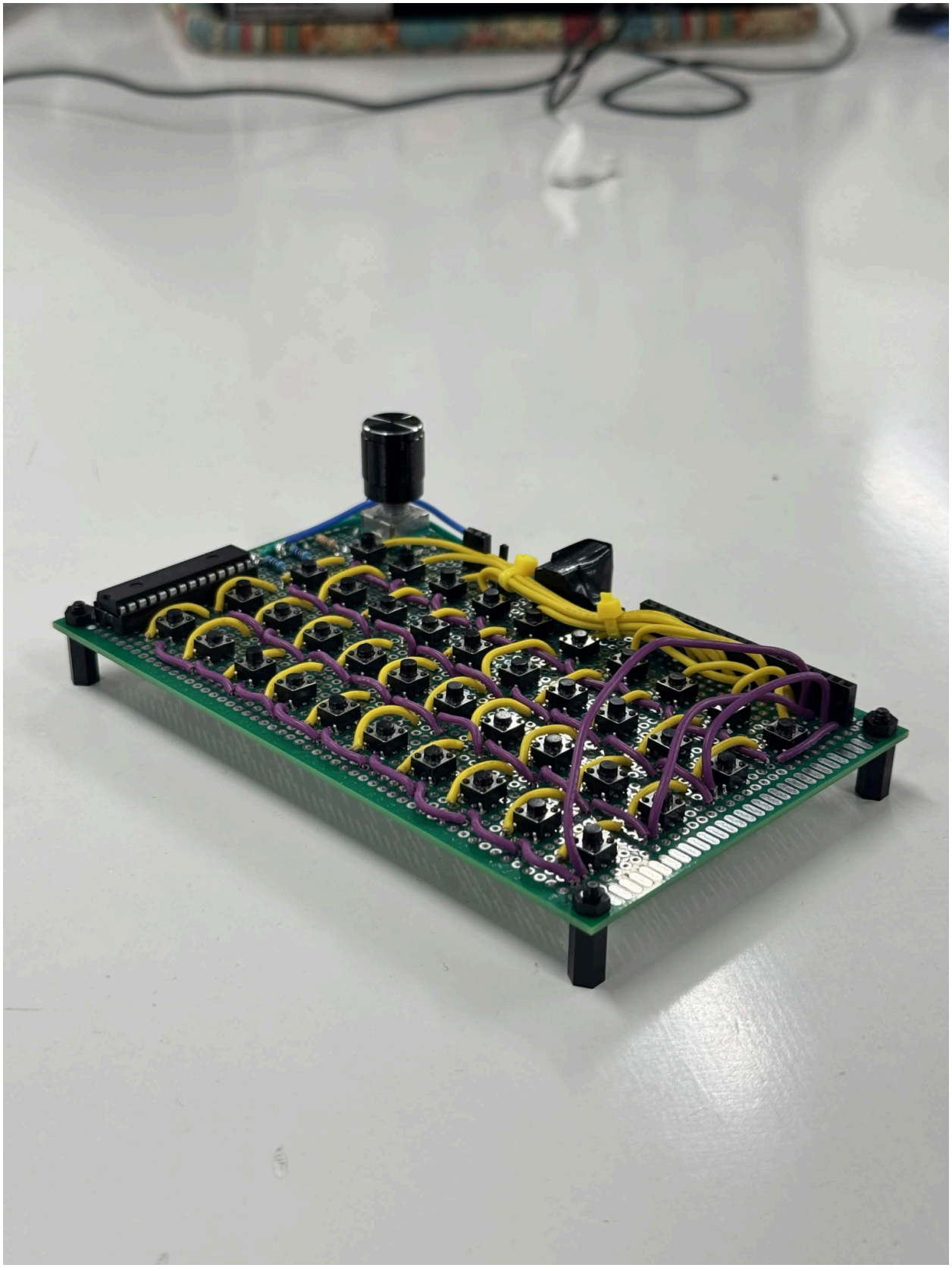


First mini prototype to validate the i2c and rotary

GPIO -> I2C Rotary encoder.



We use an I2c I/O expander to change the rotary encoder to I2c. It's a MCP23017. This tutorial helped a lot in understanding how to use it:
https://www.g7smy.co.uk/2014/08/rotary-encoders-on-the-i2c-bus/ .
The diagram of the keyboard pinout must therefore be changed because the pins have changed for the rotary encoder.



PINOUT :

Keyboard

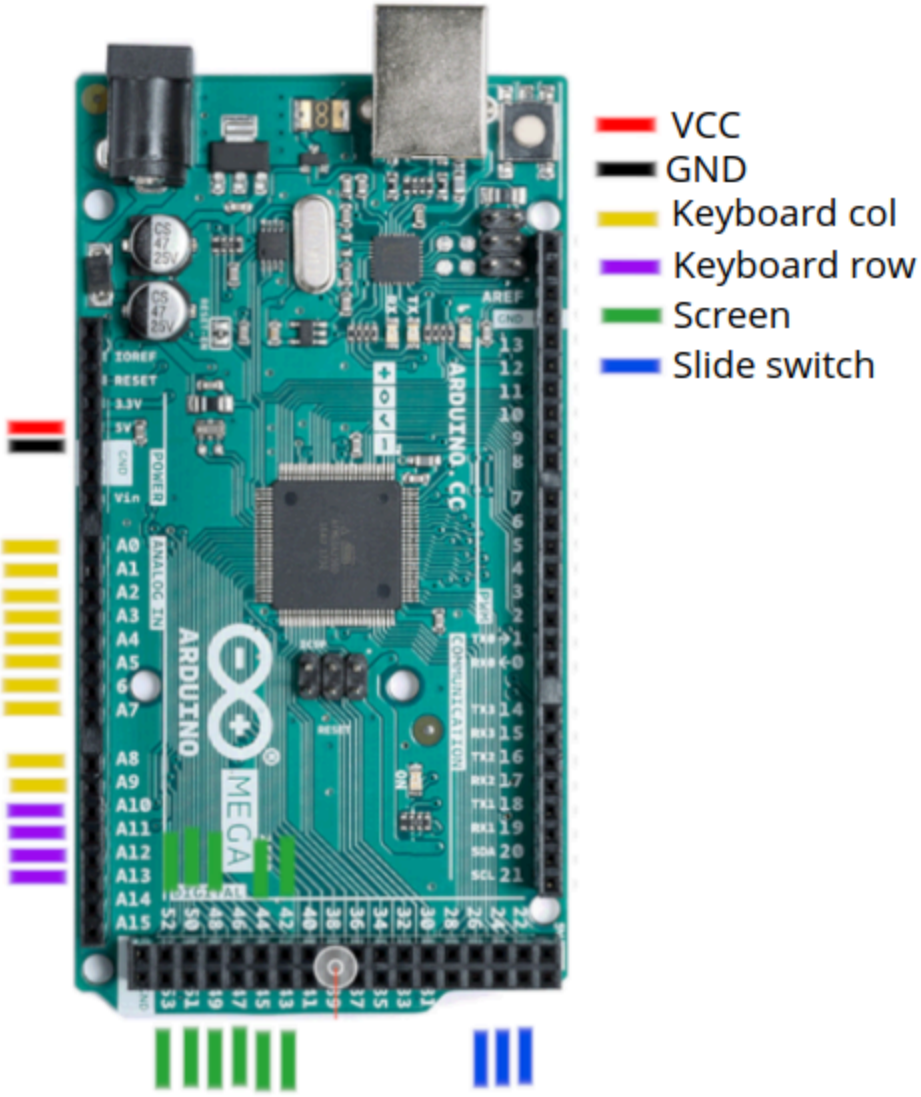
Pin Arduino	Registre	Fonction
A0	PF0	COLONNE 1
A1	PF1	COLONNE 2
A2	PF2	COLONNE 3
A3	PF3	COLONNE 4
A4	PF4	COLONNE 5
A5	PF5	COLONNE 6
A6	PF6	COLONNE 7
A7	PF7	COLONNE 8
A8	PK0	COLONNE 9
A9	PK1	COLONNE 10
A10	PK2	LIGNE 1
A11	PK3	LIGNE 2
A12	PK4	LIGNE 3
A13	PK5	LIGNE 4
22	PA0	led 1
24	PA2	led 2
26	PA4	switch pin

It should be noted that the pinout here is different from the one provided in the CDC; indeed, the pins we needed are not used in the dev board we borrowed from the lab, so we had to adapt.

Screens and SD card

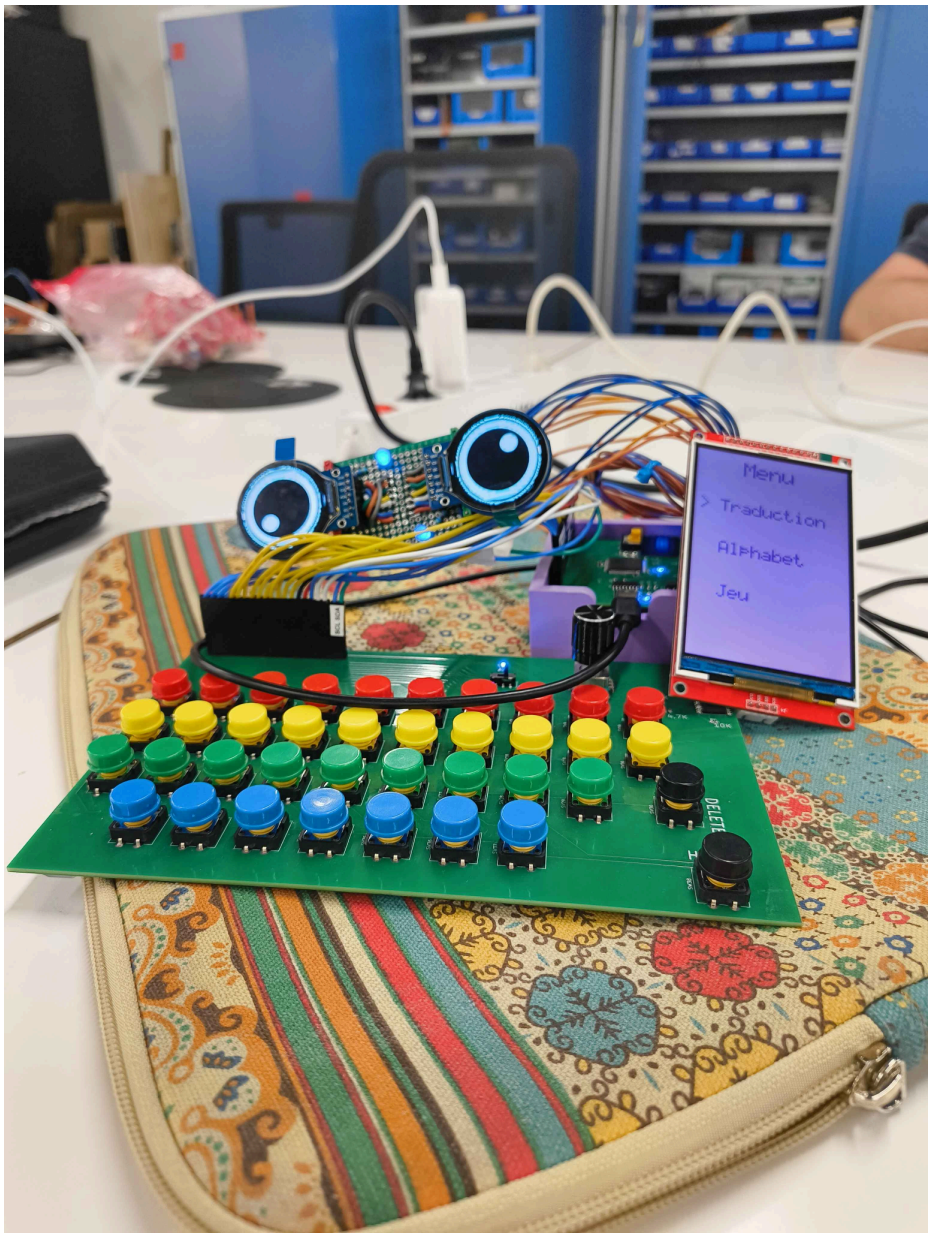
Pin Arduino	Registre	Proto Pin Nbr	Fonction
52	PB1	1	CLK
51	PB2	2	MOSI/SDA
45	PL4	3	DC
48	PL1	4	Select LCD 2
43	PL6	5	Select LCD 3
49	PL0	6	Select LCD 1
47	PL2	8	Reset LCD 2 / 3
42	PL7	7	Reset LCD 1
50	PB3	10	MISO (used for SD Card)
53	PB0 - declare in SPI INIT()	No Pin	SS - DO NOT USE
44 (check if works)	PL5	9	Card reader Select CS

LCD 1 (main screen)
LCD 2 (oeil gauche)
LCD 3 (oeil droite)


VCC
GND
Keyboard col
Keyboard row
Screen
Slide switch
ARDUINO
MEGA
ARDUINO.CC

Prototype PCB

Prototype with the first PCBs connected to our main PCB with screens.



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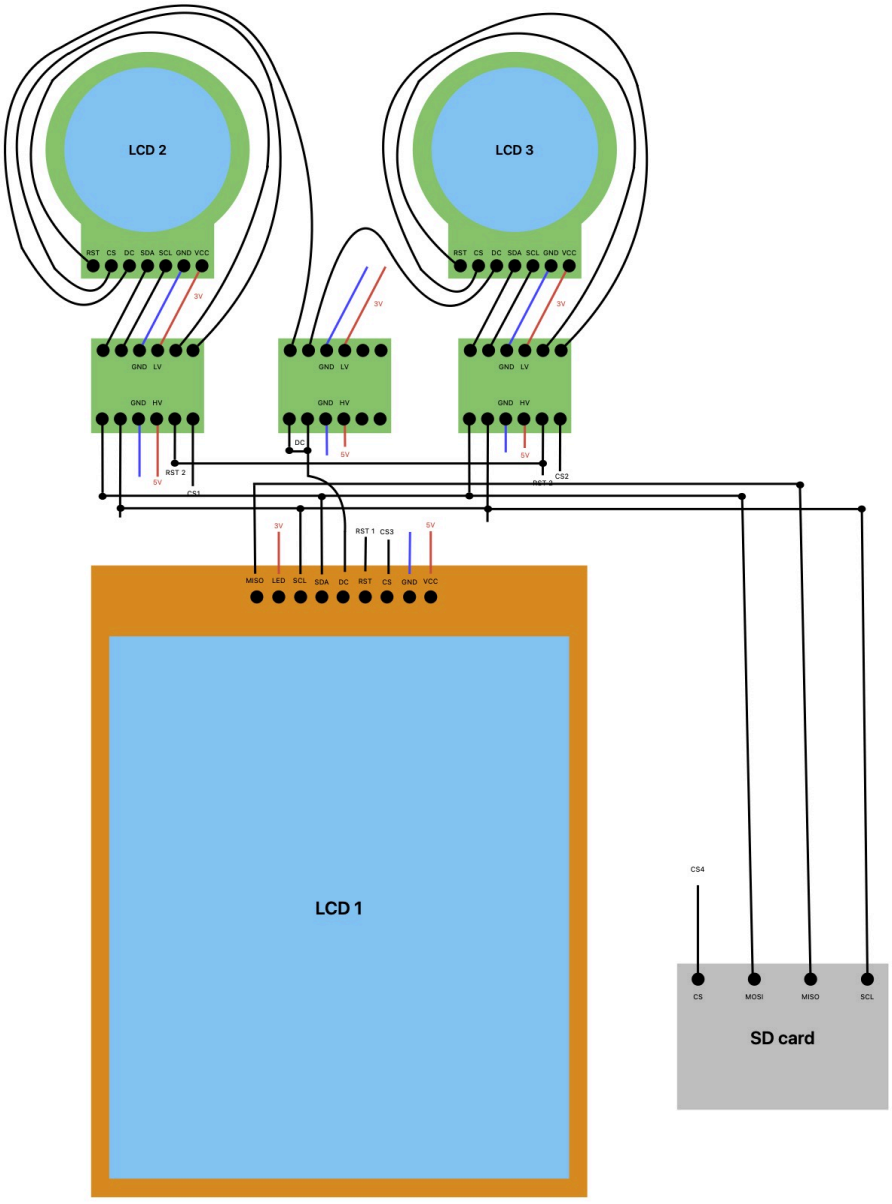
Screen


LCD 2
LCD 3
RST CS DC SDA SCL GND VCC
3V
GND LV
GND HV
5V
RST 2
CS1
CS2
DC
RST 1
CS3
MISO LED SCL SDA DC RST CS GND VCC
LCD 1
CS4
CS MOSI MISO SCL
SD card

Explanation of the schema

This diagram shows how to connect 4 devices (3 screens + 1 SD card) on the same SPI bus, with level shifters between 5 V and 3.3 V signals.

The microcontroller is not featured in this diagram, but the wires are coming from below.

Components

LCD 1: the large rectangular screen (type TFT ILI9488, with MISO, LED, SCL, SDA, DC, RST, CS, GND, VCC). It is powered by 3.3V and 5V.

LCD 2 and LCD 3: the two round screens (type GC9A01, with RST, CS, DC, SDA, SCL, GND, VCC). They operate at 3.3V, so their inputs do not support 5V signals.

3 small green cards: the level converters. On the HV (High Voltage) side, we find the 5 V and the microcontroller signals. On the LV (Low Voltage) side, there is the 3.3 V ("3V" red wire) and the signals to the round screens.

SD card: it uses CS, MOSI, MISO and SCL.

The principle of wiring

Shared lines (SPI bus): the clock (SCL), data (SDA = MOSI) and DC lines are common to all devices. These are the long horizontal black wires at the bottom. MISO is connected to the SD card and to LCD 1, the only devices that return data.

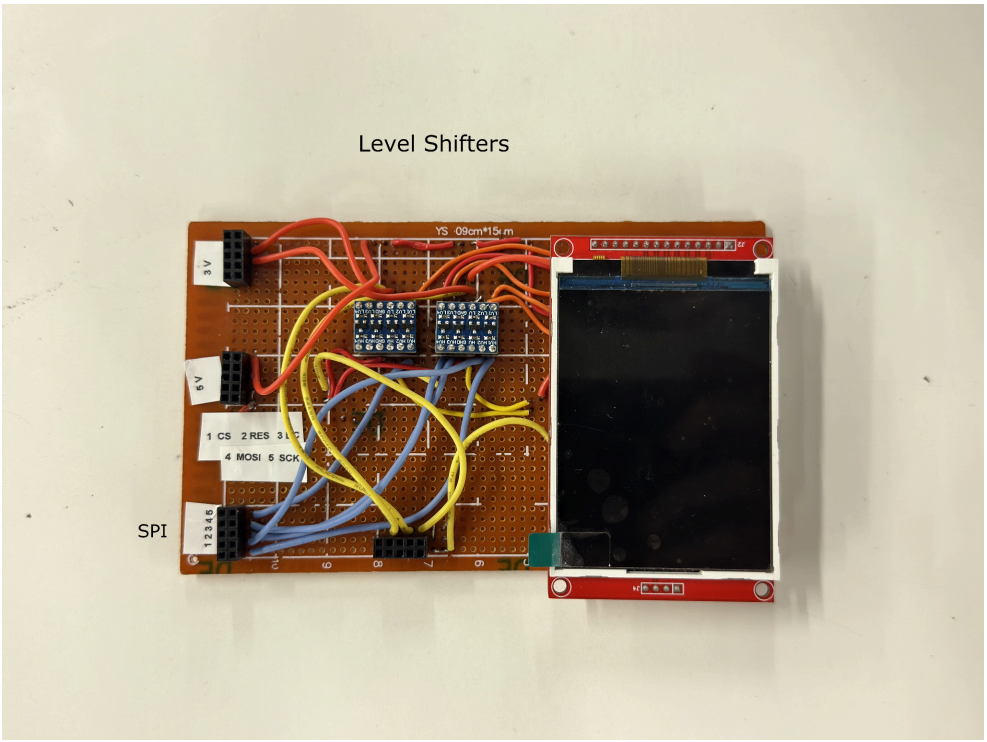
Individual lines: each device has its own CS (Chip Select) to be chosen in turn, and the screens each have their RST (hardware reset):

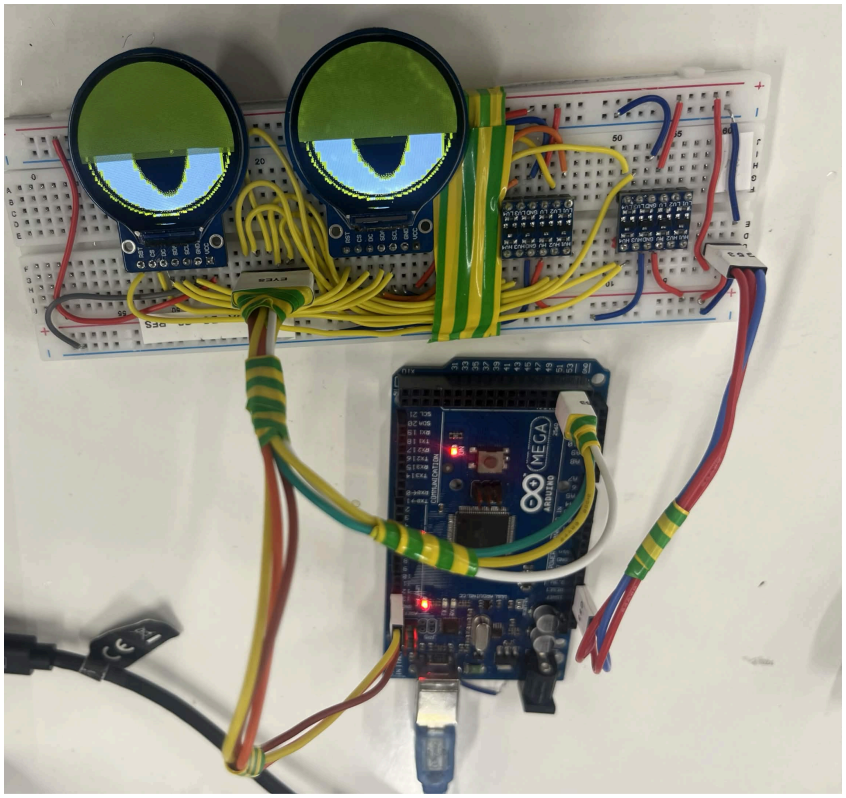
Signal	Peripheral
CS1 / RST1	LCD 1
CS2 / RST2	LCD 2
CS3 / RST2	LCD 3
CS4	SD card

Role of the converters: the 5 V signals from the microcontroller enter on the HV side, are lowered to 3.3 V on the LV side, then join the pins of the round screens. This avoids damaging the 3.3V screens. The middle converter is mainly used to connect the common DC line to the two round screens.

Some beginner’s mistakes

- The diagram does not show the power supply for the SD card (VCC/GND missing). It needs to be connected, checking if the module accepts 5V or only 3.3V.
- If the SD module does not have its own regulator and level converter, its signals must also pass through a level shifter.
- Only one device must have its CS low at a time, otherwise the data will conflict.
- We didn’t account for the possible crosstalk in this diagram and since the wires are very intertwined it should be carefully addressed.



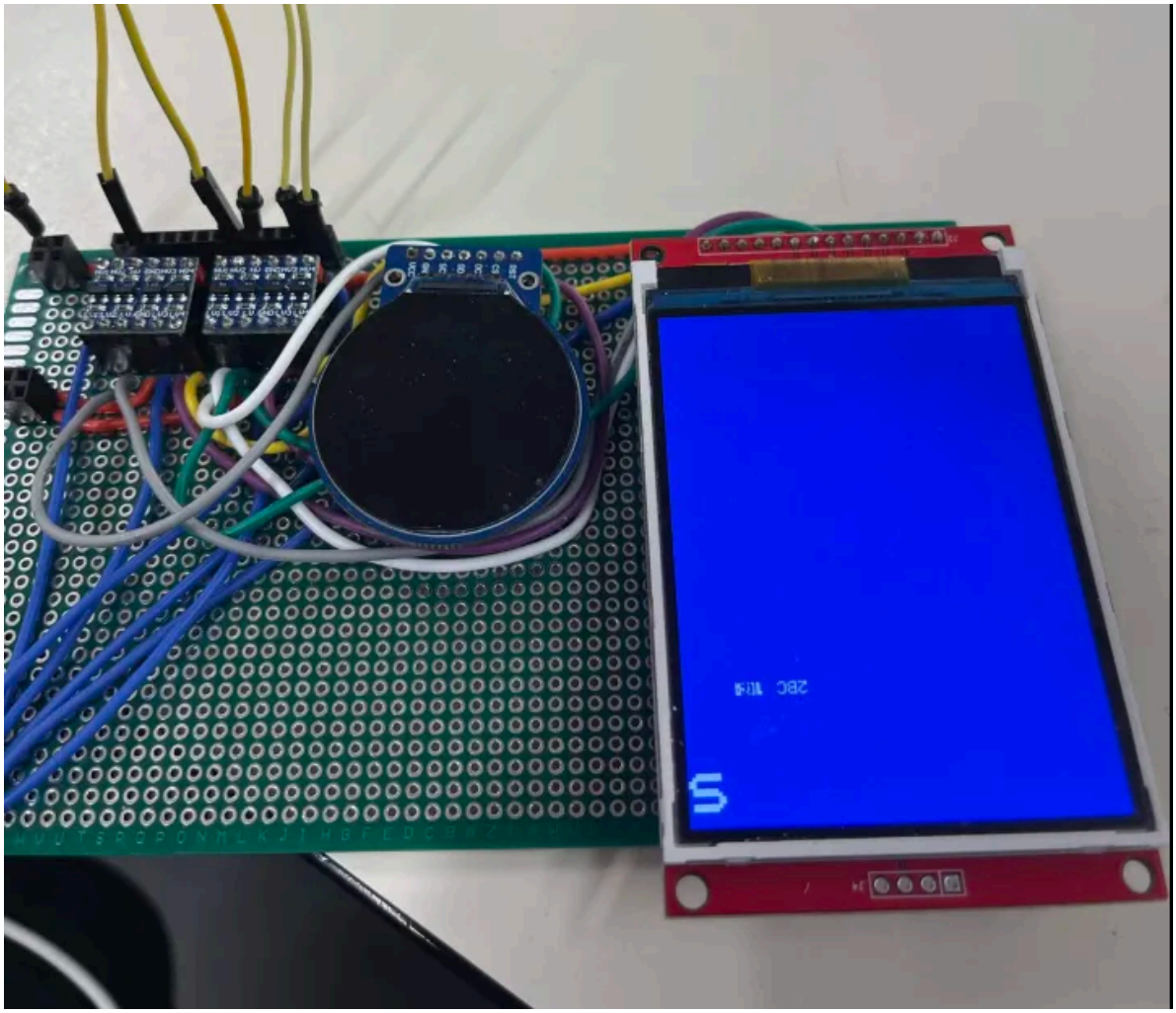


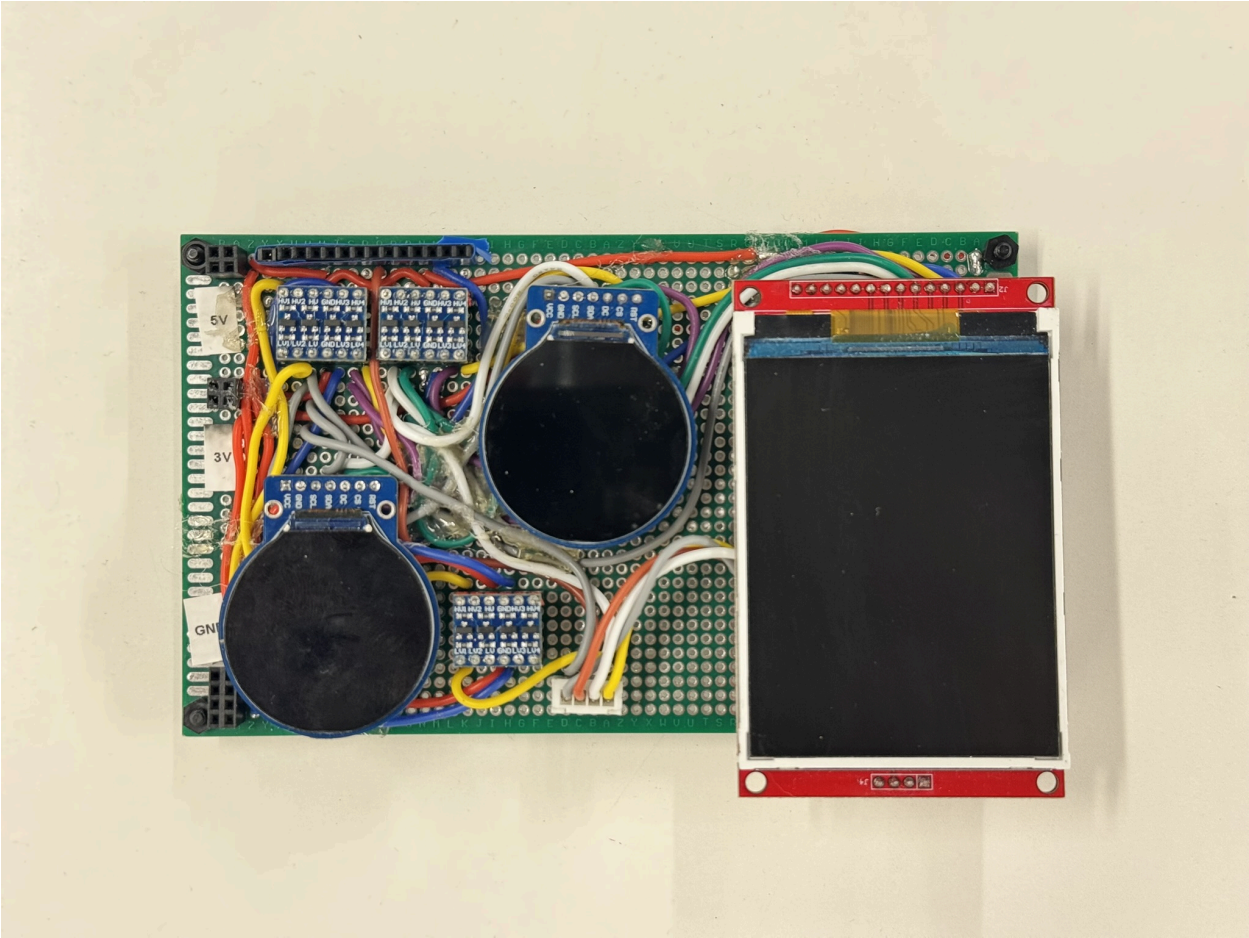
The first prototype breadboards and hand made circuits for screens

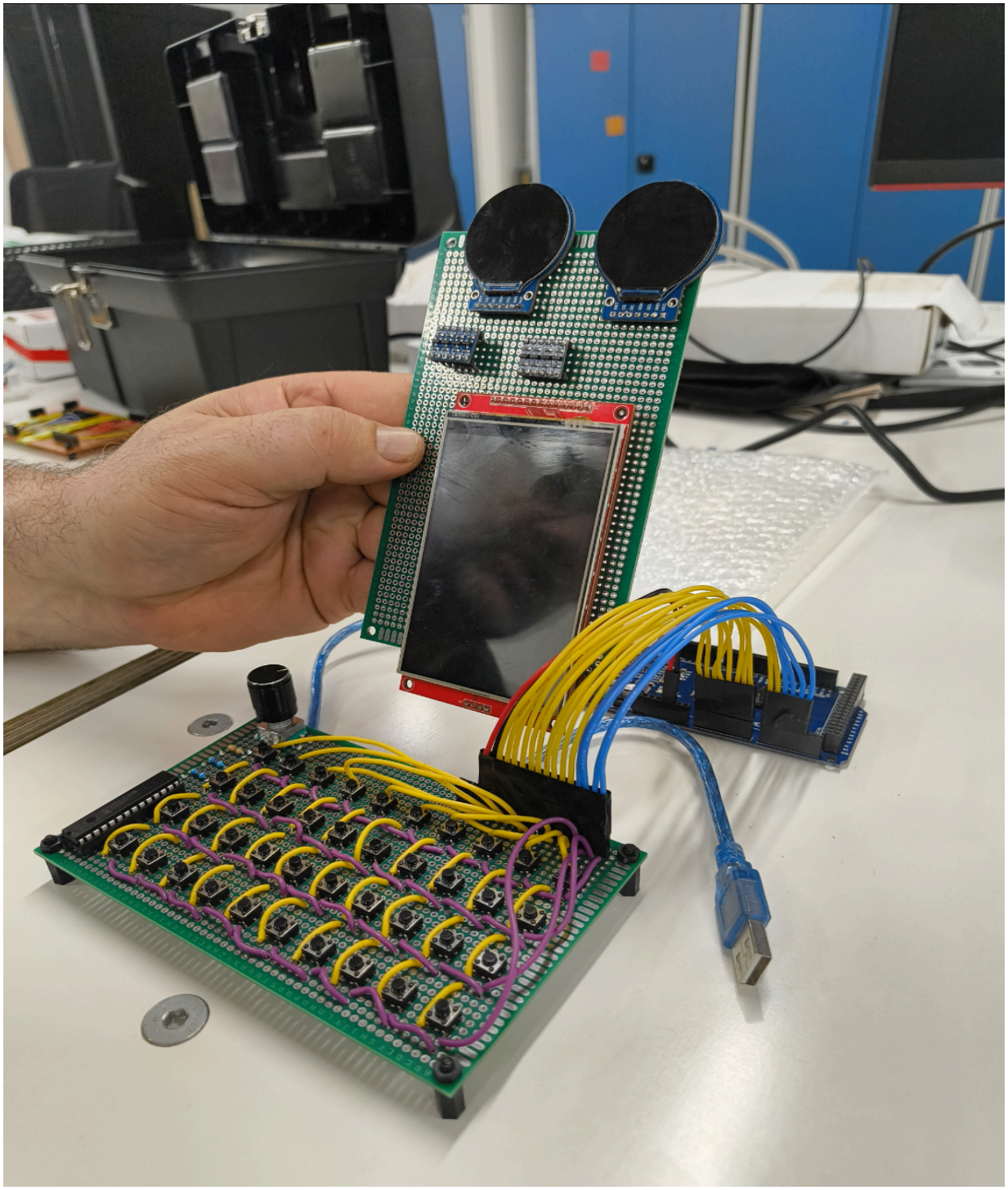
For each type of screen, we made a partial circuit in order to test the SPI communication, the initialization of the screen and the display of images and text.

A simple Arduino Mega board was enough to control the screens. It allowed us to test the scaling of the text on the main screen, as well as the filling of an area with a solid color (fill).

In a second step, we designed a single circuit combining the three LCD screens, see photo below. This was done on simple prototyping circuit board hole grid

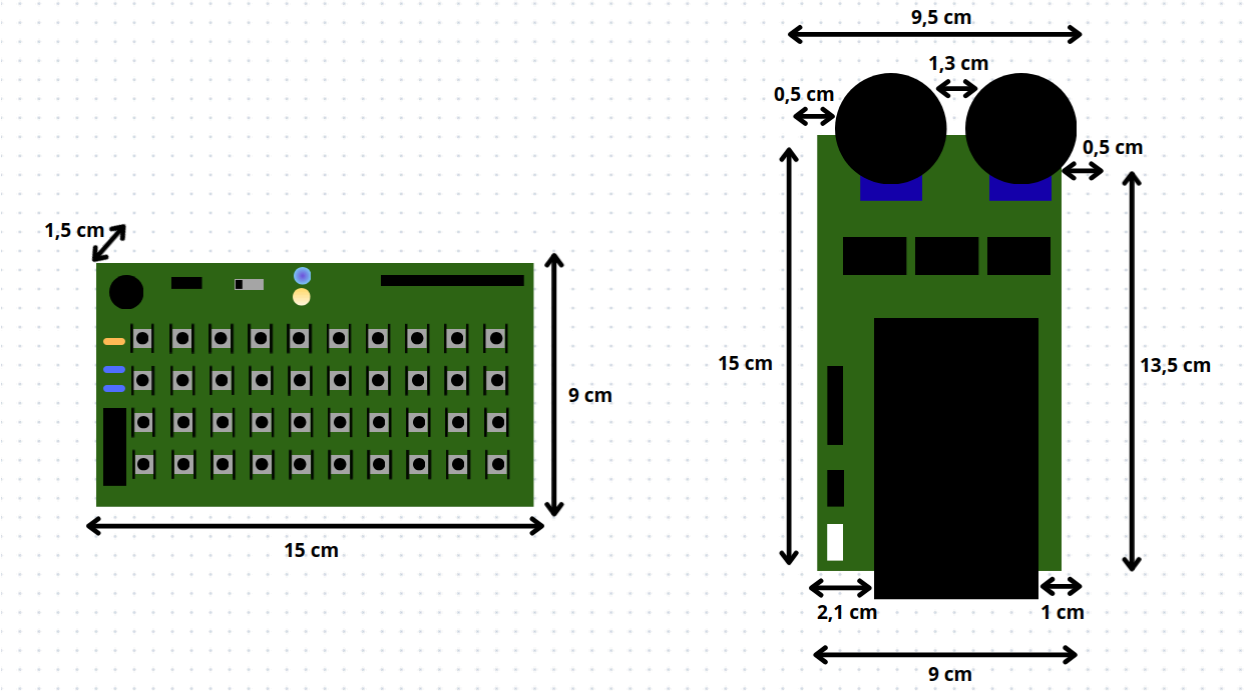



5V
3V
GN



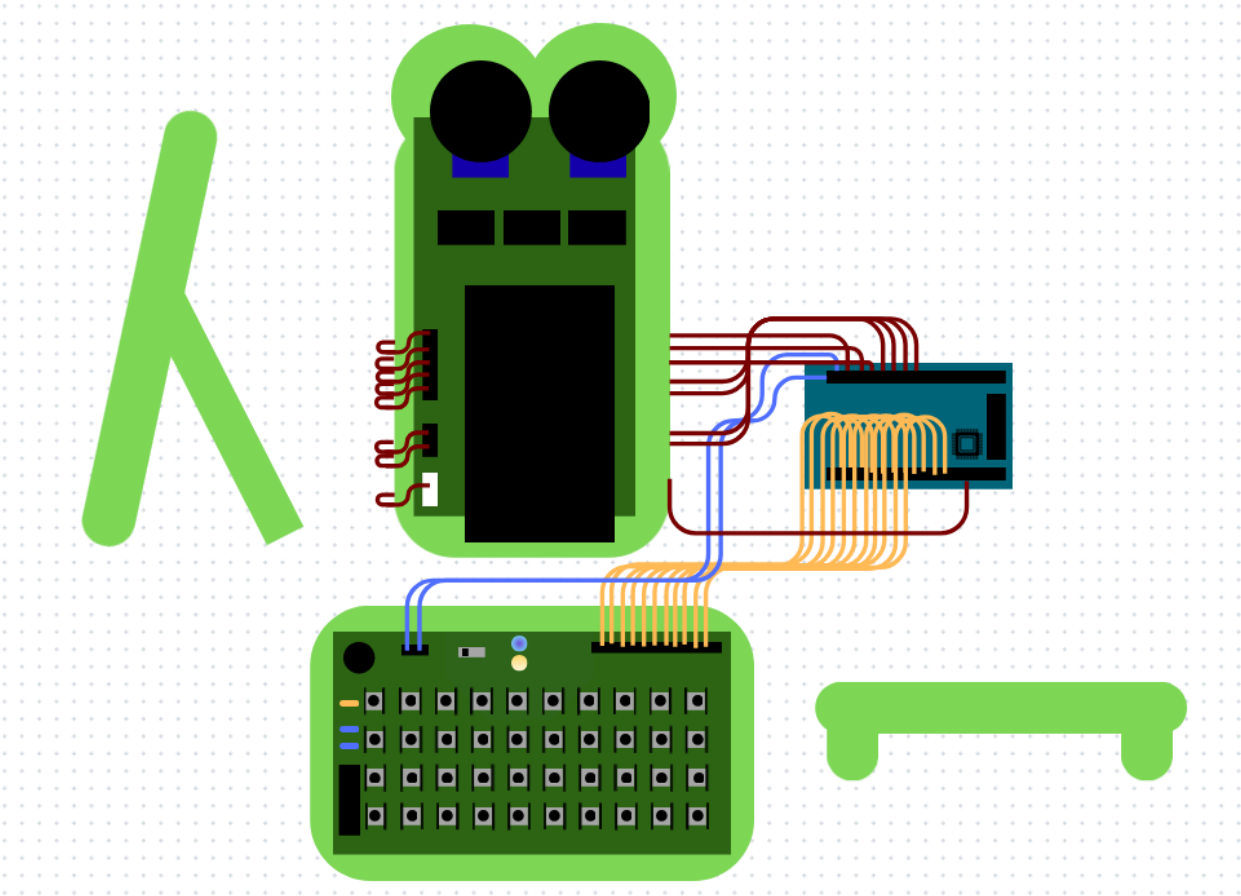
We then grouped all the elements on two circuits in order to test them.

Measures for the case:



on the first prototypes, 10/07/2026

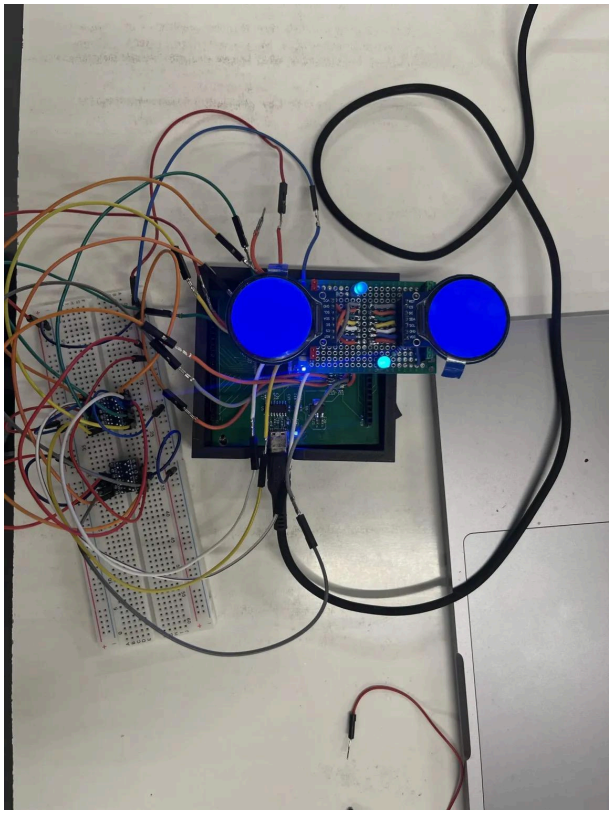
First test case:

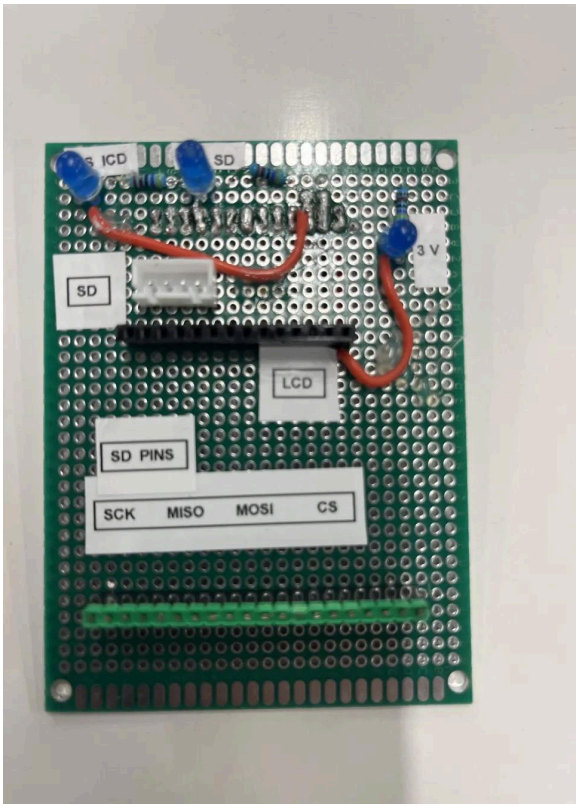


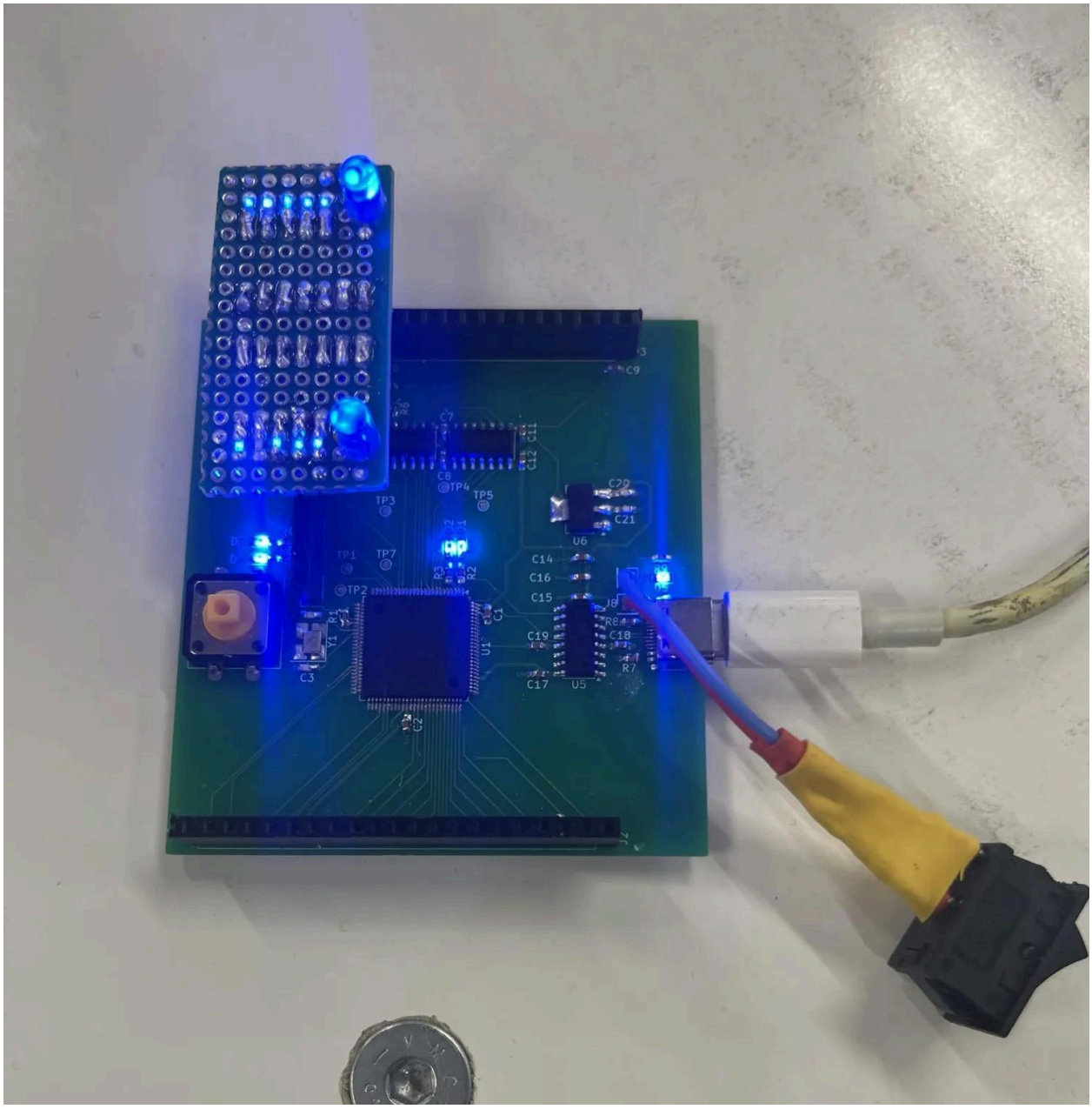
First 3D printing of the test case, it’s not really a frog but it doesn’t matter.

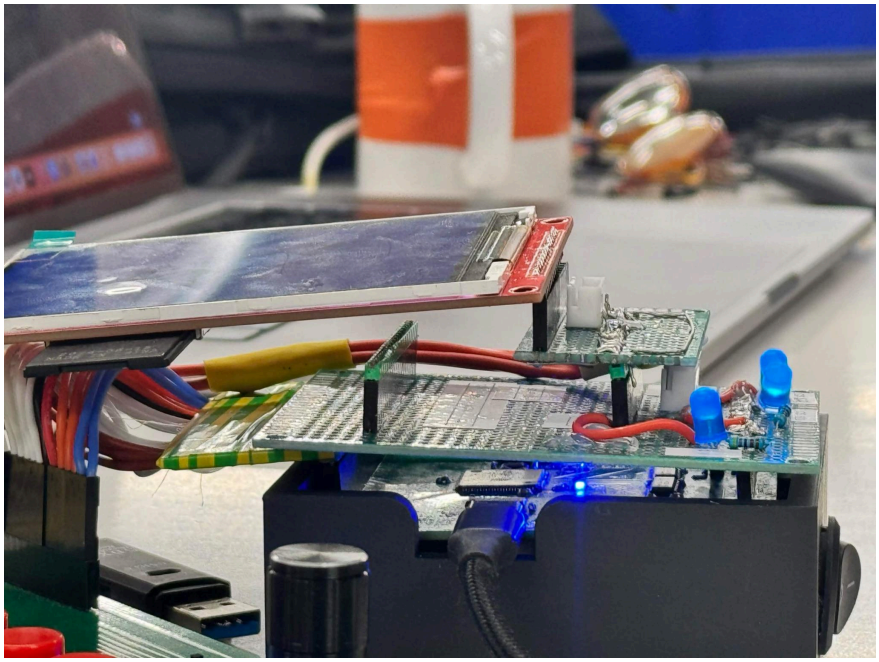
Prototype extras:

To help us test and analyze the circuits, we also manufactured adapters and diagnostic tools, such as a pin tester for screens, etc.

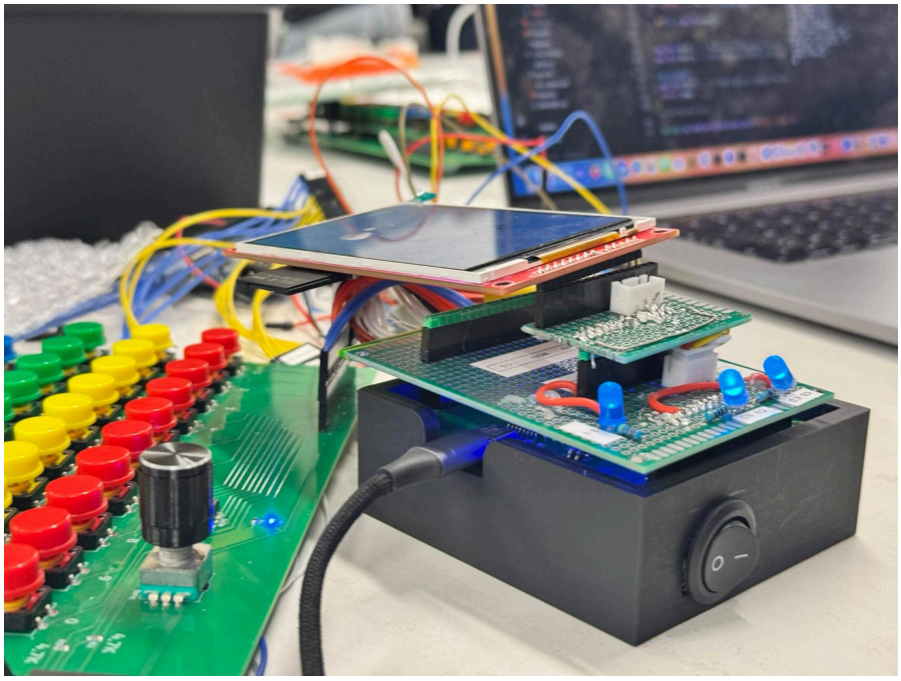





C9
C7
C11
C12
C8
TP4
TP5
TP3
C22
C21
U6
TP1
TP7
C14
C16
C15
TP2
R3
R2
C1
R1
Y1
U1
C19
C18
R8
R7
C3
C17
U5
C2



We even tried stacking our adapters, and the funny thing is that sometimes it worked, we had no idea why but it worked. It took a while to figure out that one of our major problems was crosstalk between the cables.



Our first Kicad design and printed PCB

When the PCB version 1 arrived we designed a little 3D box to protect the back of the pcb. And we had to solder wires to inverse TX and RX that had not been inverted correctly.

We also harvested some components by desoldering old PCBs in the recuperation bin of the Labelec. Thus some ATmega2560 and other parts were recuperated.

This enabled us to produce 3 working PCB version 1. Which was great help for each member of the team. This meant that we could develop the code and test it at the same time.